

ActInf GuestStream #005.1 ~ Martin Biehl

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great hello and welcome everyone to actin flab today is april 28 2021 it's actin flab guest stream number 5.1 with martin beale martin thanks so much for joining us on today's stream and we're really looking forward to hearing about your perspective and what you have to share here so uh if anyone in the live chat would like to ask questions we'll be relaying those to you during and at the end of the stream other than that please take it away and thanks again for coming on uh thank you daniel um welcome to my talk everybody uh i will talk about uh or i will talk about at least in the background about the paper i wrote with felix pollock and rio techni which is called technical critique of some parts of the free energy principle but i will not go directly through this paper i will kind of try to give a more updated perspective um yeah what else to say i work at uh aurier in tokyo and this work is also funded by temperature world charity foundation and fedex was supported by the monash university okay so here's the content of this talk first i'll make some preliminary remarks and then then talk about my perspective basically on the free energy principle and how it can maybe maybe split into two parts uh the construction of agents and the identification of agents and then talk about what the fep agent construction is and what the fvp agent identification method kind of is and then i'll talk about the relation between those or at least the claimed relation between those and then i'll go through some issues with this agenda identification method and also with this relation okay um so yeah like i said the main publication uh is same name as the has the same title as the stock and um came out this year and it resulted from an effort with uh felix where we we actually set out to do like a discretized uh version of the life as we know it paper from 2013 but we failed and then in the end we wrote this critique um so yeah i will i will give this a in this talk i will give a my personal view a little bit of the free energy principle and why i find it interesting and like i said i want to go through the paper in detail um and instead summarize uh how i currently see the issues also the issues that we mentioned in the paper um but the ffp the free so fvp meets free as while we were writing this paper it has changed and so my perspective has also changed and but uh if you have questions uh particular questions uh about the paper then you can of course ask me afterwards [Music] maybe a little bit more detailed

overview so in my opinion the free energy principle can be viewed as two parts um and the first part is basically what uh maybe you've heard this kind of this statement that markov blanket implies uh markov blankets imply bayesian inference and i so from my perspective this is kind of a way to identify agents in stochastic processes so this is one part that is contained in the fep literature and another part is about how to construct artificial agents for for example for reinforcement learning or partially or observable markov decision problems and i call this the constructing agents part and the claim the claim is that uh like if you read papers about this constructing agent pod the claim is that they that this construction is based on this identifying agent spot so the construction is based on this markov blanket's employability and inference um part of the literature and i will kind of argue that this is has not been established and i also kind of think it's questionable hey at least in the current state then i also present some other issues with the fep among those those that i mentioned in the paper and these issues so yeah one thing i want to say is that these issues they kind of affect would affect i mean if if they are if i'm if i'm correct about those issues then they affect the usefulness of the free energy or agent identification so so then we couldn't uh identify agents using the fep and this also would question it as a theory of what life is or of life general or life as we know it if you want and then um it would also affect this claim that uh these three these constructed free energy principal agents are somehow based on first principles more than let's say some other reinforcement learning agents what uh what all of this what all of these points in my opinion won't affect is um that free that these free energy uh principal agents like free energy minimizing agents that are constructed in the literature that they work or uh that they work at all basically or that they can be useful for some research i mean i think i think those i think that these agents work is uh kind of independent of this whole markov blanket implies basically inference part but maybe i'm wrong but okay so what's this uh what are these two things uh in the construction of agents and the identification of agents so like i said i think the literature can be split into these two parts and like how to construct an agent or a controller and the other one is how to identify basically an agent in a stochastic dynamical system and yeah so and like i said also before the the this first method how to construct um agents or controller is claimed to inherit from the second method and these are still claims that are uh current so there's a paper that came out just this year or actually a week ago so i think it's called sophisticated inference okay so what do i mean by construction of agents or what is this uh what is the problem of construction of agents um because that's a yeah a different problem than the identification of ages that's kind of the first point i want to make at least two um

problems are different in general so yeah in artificial intelligence research what what people are occupied with uh is the construction of intelligent agents for some given environment or task or like um and uh yeah so basically what what these people do in artificial intelligence research or engineering is programming robots or virtual agents and of course the robot is then kind of fixed a fixed environment or if you have a virtual world then this is also a fixed environment and then if you have some reinforcement learning problem then you have kind of a task okay so um like i said we in these cases what we have is some environment uh that is given and i uh i write this here as a bayesian network and then we construct a controller and then this what this basically does is we have some environment states that i call e and sensor values that depend on each other and we have dynamics of each other so so basically towards the right the time increases time step increases um so we have uh we have some dynamics of this environment which tells us how each changes under the influence of actions and depending on its previous state and the environment produces sensor values and what an agent does is it takes these sensor values and updates its uh internal state which is called μ and produces actions accordingly so as you see here μ depends on both the sensor values and the actions how exactly these edges are look there's some variability in it but it's roughly one one idea like yeah most of them are roughly equivalent so like i said or a bit more formally maybe uh each of t is an environment state or an environment or a task is defined as by an environment state sensor values some kind of action interface so some dependency on the external variable so this dependency of the update of μ on the action and the dynamics which means if we have a if we are in discrete time then the dynamics can be specified as markov kernel or conditional probability of the next or the yeah the next internal state the next sensor produces sensor value conditioned on the previous internal past sensor value and the action which is the influence from the patient and in continuous time we could also formulate this and one way to formulate this looks like this we give the uh time derivative of the internal state as some function on its own uh state and the sensor and the action plus some noise and similarly for the sensor dynamics and maybe it's not so important but in general you can have and an environment can have some kind of reward that it gives to the agent as part of the sensor value or um it may not okay and then what's an agent so an agent has this uh internal or memory state and it produces actions and it also has an interface for accepting or sensors that are produced by the environment and its dynamics this can also be written as this kind of conditional probability or similarly with time derivatives in the time continuous state and maybe the the main part uh that's going to be relevant for some something later in this talk is

that we can combine the we can combine such environments and uh agents and then they form a markov process so then we have uh some joint variable x and the product of these two uh conditional probabilities the one for the environment and the one for the agent they induce this um markov kernel of the joint process and similarly in the continuous case we can uh see the whole thing as one the combination of the environment and the agent as all as a single uh very as a single markov process with uh state x okay so that's kind of what we maybe should keep in mind um some remarks uh on this usually we have we construct these agents to solve some problems so we have some kind of reward um so this is the case in reinforcement learning for po mdps but we can also have agents that have just intrinsic goals and they don't need any reward from the environment these things are usually called intrinsic motivations and the these the fep agents from the literature they actually kind of have both uh in within them so they have a they contain some part that's like an intrinsic motivation and they also can um they have some part that is similar to a reward and so basically the entrance motivation part is is like a maximization of information gain and the uh instead of reward maximization we minimize some divergence from uh a desired distribution which is good but at least on a high level this can be seen as uh so these two things are kind of combined in the free energy in the expected free energy yeah i mean there are some tricks to this but roughly this is true so that that's that's what uh i i hope it came across uh what i mean by construction of agents i don't know if there were any questions uh already and it's not yet continue thank you not yet okay so then uh what do i mean by identification of agents which is maybe a bit less uh common practice so first maybe let's ask the question what what is an agent uh in general um and uh so very roughly there's a definition by uh and they say um to be an agent you have to be something that's individuated that can act and make things happen and can act which just means to make things happen basically and uh it can act to achieve the these actions are are performed in order to achieve some goal so an agent has also a goal maybe something that has a goal an x to achieve actively pursues this and these these agents that i just talked about these constructed agents that we construct in artificial but also in the fvp literature they are agents because they are kind of by construction they are individuated because we kind of know where their boundary is uh like if you ask some engineer what they have to do to construct the agent they know exactly what to do so they they seem to know where the agent uh is or what the agent is and of course since we program into them uh the intrinsic motivation or some kind of reward maximization mechanism they are also co-directed um okay so but then this is not this is just what what agents are so what's what's what about identification of so the the at least for

me the origin of this question is uh the kind of the problem of agents in physics so the physicists they tell us that everything is uh determined by the laws of physics so that means everything that happens is made happen by the laws of physics and not by any thing within this uh within the universe so in this if they are correct then nothing within the universe can make anything happen but that also means that nothing can act to achieve a goal because nothing can make anything happen so you can't act to achieve some goal because you can't act in the first place so this seems to suggest that basically there are no agents in the universe like it must be some kind of illusion at the same time i mean we are here uh we are agents and we kind of want to we kind of have goals or at least it seems extremely likely and it also seems like there are things that don't and we built these robots and they seem to be quite real things and um have goals also so it's kind of so maybe there is some way there should be some way to make these two things compatible um so what we want is uh at least if you're if you like the uh physicist's perspective um and also you like you like to think about agents then you maybe you what you want is a formal theory of agents that is compatible with physics and then so one one problem that such any such theory should be able to solve is that if you have some kind of universe and in physics that's basically just a dynamical system uh then what and where are the agents so if you have if you have a formal definition of agents or former theory of agents then this is the question that it should probably be able to answer so here's the universe um and what we want to do is we want to check where inside this long history of the universe or uh there are these things that are maybe can be called agents like animal here a bit more formally so we have some kind of in physics we have some kind of dynamical system we have a state space and states and we have a dynamical law that is looked maybe looks like this uh the time derivative of y is some function of the so the time derivative of the state of the universe is a function of the current state of the and then we have so these things have trajectories and if we select one of those trajectories by some initial condition y_0 then what we would expect from a theory about a theory about agents and one that can also identify agents then what we would want is that this theory gives us like once we give it such a trajectory of the dynamical system or maybe the trajectory and the that this theory tells us what is the set of agents contained in this trajectory and it should tell us this including what is basically its body or its individual the what is the individual uh what are the actions and what are the goals okay so this is what i this is kind of the agent identification problem maybe and uh the fvp um does something like this they start not directly maybe with a deterministic dynamical system as the physicists maybe like but they instead start with something that's usually considered as a subsystem of such a

deterministic universe and this subsystem should obey this kind of longing equation $\dot{U}$ and then the free energy principle tells us what the conditions under which there's an agent inside such a thing so I'm not 100% sure how this relates to the exact formulation I gave here where you just get a trajectory and the agent theory tells you where the agents are there are some subtleties to this but roughly I think the free energy principle is definitely a step towards such a theory and this part of the free energy is what I call the free energy principle identification the free energy agent identification method and it is if you're familiar with the literature then this is where the Markov blanket placed the fundamental the so-called mark of nucleation maybe say okay Martin can I ask $\dot{U}$ what do we gain or lose by going to the lying given equation what is what is gained or lost in that maneuver relative to the more general phrasing you had on the previous slide yeah good question so I well I I don't really know $\dot{U}$ I all I can say is that yeah I mean one thing you definitely gain is that if you I mean this thing it's kind of uh hopeless like if you would really uh if you if we would need the particle equation then we would probably never be able to identify uh agents because the particle I mean it's just too complicated and complex uh to say much about it but so here this gives us kind of we can kind of say okay I can ignore a lot of stuff and all I have to find is some kind of part of the system that uh obeys some kind of approximate uh equation basically and this can be enough $\dot{U}$ so this is I guess this is one advantage yeah I mean like there's one other thing I'm not sure I mean there's one other thing I sometimes think uh where this is might be useful for and that's so what we what we kind of want $\dot{U}$ is that this agent theory I mean if you think about it uh basically a billion year like a few at some point there probably weren't any agents inside the universe so so at that point uh so so agents kind of uh like a full-blown theory of agents they it should allow trajectories with uh within which agents appear and basically disappear or at least they have to be able to appear because that's kind of what we think the happened in the universe and but this is not $\dot{U}$ this kind of appearance and disappearance is actually not so easy to do especially if you would just kind of naively split up this the whole system into two parts and say is one part the agent and one the environment because then you split it for all time and this is kind of weird uh because how because you you would want that the split only works for a short time or or once they actually uh is an agent so sometimes I think okay maybe maybe these this approximate equation only works for a short time and that's an advantage of this also uh there might be short time frames where this equation works and where they and others where it doesn't work and then we would get uh we could think about whether this means $\dot{U}$ the agent appears or not but actually I'm not so sure what these $\dot{U}$ and

what the details so yeah that's all i can really say this i mean i think yeah it's uh but it's a very good question i think thanks okay uh any other questions because it would be a reasonable point uh not yet but anyone is welcome to do so thank you though okay um so how does the fep construct agents i'm not gonna talk about this a lot but uh just maybe i'm only going to maybe mention the things that are relevant later and this is basically just the discrete time free energy minimizing agents so in the free energy friends in the fep especially the early ones you also have continuous time free energy minimizing agents but uh yeah i'm gonna mostly ignore them here they're not really there's nothing in particular i want to say about them but um so in this discrete time free energy the in this discrete time fep agents the agent state μ consists of hyper parameters um basically hyper parameters for the conjugate priors of the models that this agent is constructed to uh use basically and these hyper parameters they encode the beliefs or um both the environment state the basically a model of the environment and a model of the dynamics sometimes there's no model of the environment dynamics but sometimes so that's kind of important i mean they basically yeah so the actual state uh consists of these hyper parameters um and then what happens uh what happens with this is that uh in response to some observation and past action the agent updates the beliefs and actually that just means that it updates these hyper parameters and this is done via variation of base and um then the agent basically uh it also has a belief over actions and this also gets updated and this is done in in such a way that it minimizes the expected free energy which i think i've mentioned this before maximizes information gain and kind of minimizes the divergence from desired release um maybe somewhat relevant is that uh if the model class that the agent maintains contains the true basically the true or the truth the true model then um or an exact model of the environment and so on then the belief update this variational belief update actually is the same as the exact bayesian update okay so that's already all i say about the fvp agent construction and now i'll talk a bit about how the fep agent identification works so the the main relevant literature on this uh agent identification are four papers as far as i can see and the first one is live as we know it from 2013 and then there's the monograph uh quite a long paper i think 180 pages something or 80 page i don't know lots of pages and that's called the free energy principle for a particular then there's a kind of an extract from that but with small differences and it's called markov blankets information geometry and stochastic thermodynamics by thomas parr and then there's also a paper which is written in response to our criticism that's called some interesting observations on the free energy principle say a question nope sorry okay so uh okay so this this last paper was in response to our criticism so these i think these

contain basically what these contain all the things relevant to this agent identification problem um and basically they are the main papers concerning this mark of naked thing there are also some other ones but i think actually these ones are the essential ones okay uh yeah our the paper we wrote it's mainly concerned with life as we know it because the second none of the other papers existed when we started writing it uh the this one we energy principle for particular physics came out while while we were maybe yeah while we were writing it um but yeah as i said i i will try i will try to um give a uh my perspective that also includes uh these newer constructions or approaches or papers okay yeah so what is the rough idea of this i can't go through this in detail mainly or among other re the reasons for this is that i don't really know i don't really understand how it works in detail like i i'm still not still haven't gone through the whole construction um because i get usually i get just stuck before i'm through okay um the rough idea is so we have our universe and like i said we find this uh subsystem that obeys the language equation um omega is some kind of markov and white noise and we also require that this subsystem or yeah is such that not only does it obey this equation but also that this has a steady state distribution p^* okay i think this is just going to be the same thing again we already saw this ah okay so now what we're going to do is we find the subsystem basically let's say uh we have this system and it obeys this um equation i mean so basically imagine this environment with this animal and now we further split up this into some oh actually i thought there was going to be something i'm sorry so we further split up this system basically we uh we split it up into here you you basic i hope you can kind of see that there's a central nervous system of this animal and we kind of split split up this whole system into relevant into parts that are aligned with agency in some way let's go this way okay okay now okay that's what we're gonna do afterwards in one step but first uh sorry it's a little bit confusing but there is uh one nice fact about so this should have been before these images there's one nice fact about these kind of equations sorry let me just go back here about these kind of equations if they have a steady state distribution and only if they have a steady state distribution if they don't then you can do the following so in that case if they have a steady state distribution then you can write f of x as this uh kind of gradient descent or ascent you you formulated of the log probability of the log of the stationary distribution and uh yeah so then you also get these two matrices here in front uh and r is antisymmetric and γ is symmetric um and also yeah you can write this here the the lower probability the log a state probability um you can write it or turn it into this uh negative log probability and that's the definition of this i you've maybe seen in the papers the surprise okay um yeah yeah like this is

not really trivial this when exactly you can write this it's not really trivial um especially when this gamma matrix and r matrix don't depend on the position x which is something that uh is usually assumed in the fvp yeah i mean if the force is linear or if f yeah the vector field or sometimes also called flow is linear then it works and this also means that around fixed points of a non-linear vector field it works okay can i ask you a question on that last slide martin yeah um maybe giving an example of a kind of system that people apply it to what is x here and then what does it mean that r is anti-symmetric and that gamma is okay so i actually don't know i don't think anybody applies this to anything uh let me think at least yeah i mean like that's actually one of the criticisms i have um but we can definitely apply it to some uh systems and maybe i'm gonna look somewhere deep down there oh that was i know these pictures they're slow always get trapped in those right i saw it okay yeah here okay so here's a 2d example of such an equation and with the gamma and the r so we say this false let's say this force or this vector field is linear so it's just a matrix times the vector so then exactly then we have this matrix times the state and we get this equation and we can uh find gamma and r you can compute them there's a way in the literature you can also see it in it's also mentioned in our paper how to compute those and you can and we can compute the stationary distribution it's some kind of gaussian in this case and it looks like this and you see here's the on the left you see this vector field and on the right you see the steady distribution here it's called ergodic density but that's the same thing and how do these split ups look so if you look at the component gamma times the gradient you get this vector field and if you look at the component oh this should be oh yeah r sometimes this is called q but that's the same thing just in case you're wondering if you look at the component r then it has this it's basically the circular uh component of this vector field it's hard to see that this vector field here is has this circular component but if you see roughly here you see that they actually aren't so this one is actually straight always cool so the left one is the hill climbing aspect and the right side is the is it what's sometimes called solenoidal or the iso contours yes so exactly and um so there is one interesting thing about this and that's that uh this is the so whether this matrix r is everywhere zero or not decides whether uh the steady state is a non-equilibrium steady state or an equilibrium steady state um and this is due to the i mean so the standard definition of non-equilibrium steady state and equilibrium or distinction between non-equilibrium steady state and equilibrium state is that in the equilibrium steady state you have detailed balance and in the non-equilibrium steady state you don't and if you have this solenoidal flow then you don't have you that's exactly when you don't have uh it basically destroys the detailed

balance um yeah detail yeah so detailed balance means that basically if you if you make a square small square here then the flow uh or two let's take two neighboring uh squares and then if it had detailed balance then the flow from one square to the other would always be the same as the square from the other to the first from the second to the first but since here this flow always goes in in this horizontal direction or not horizontally in this organization direction what goes in from the eye one side is never compensated uh in the other direction so then you can say hey but on the left side it also doesn't look like it's compensated and that that's that's true but it is actually compensated by the noise because this is only uh this is only the vector field and without the noise factor so this is only f of x without the noise factor and the noise factor in this because it's gaussian the noise factor is also it has no curl so uh this is why you um so the noise can compensate this this uh component but it cannot compensate all right maybe that was a bit maybe i shouldn't have taken that much time to explain this but it's kind of nice yeah where are we yeah okay uh yeah so this is what uh we get this nice decomposition and uh it's kind of needed earlier we only get it if we have this a steady uh and it plays a reasonably important role in the literature okay so this other thing that we do with our subsystem is we split it up in some way and this is kind of what the mark of the so-called markov blanket does so the but here i'm just going to i'm not going to talk about what the marketplace is yet i'm just going to say what it's uh so about the work that it's supposed to so it will give us um this split of the state into four of the state coordinates into four basically a partition of the coordinates and uh we will then call uh each of the external states as the sensory states α the active and action states and μ the internal states sorry someone else just joined was that a question oh no no someone else just joined in not actually sure but c'est la vie and i'll add a little uh password just so that no one else joins him yep go for it okay uh so so then what the mark of blanket also does basically it does all kinds of it does a lot of work let's put it this way because all you need is that you have this subsystem and you have the steady at the mark of blanket and then what you get is that the internal active and sensory states in this system appear to and now this is important actively resist the dispersion of its markov blanket okay there's this word markov flanked again so here you should read boundary or basic you can also read individuality that's what this means in this case um by engaging in approximate vision inference so so this this statement basically has all these features that we kind of want from an agent identification theory because it tells us that if we have this under these conditions and by naming these variables in this way then we see that these i mean it basically it it makes sense to name these variables in this way because they we

have um some kind of goal the goal is to resist dispersion of the mark of blankets so basically it's the goal is kind of persistence and uh the agent acts uh to pursue this goal because it actively resists this dispersion um yeah and it is individuated because we have this uh we know which states belong to it and which states basically don't belong to it if this distinction between external and internal um yeah and then additionally which is also kind of nice that this whole this agent is also doing some kind of approximate vision inference which is a bonus i guess so um this is why i i say that uh the fep basically is contains a theory of agent and um yeah and this is yeah and this is also this is basically the part that interests me the most about the it's yeah okay it's kind of uh maybe a unique selling point compared to all other kinds of uh to most other approaches to constructing artificial agents and it's also i mean there are not many there are not many agent identification theories that i know and i actually hope that i i know all of them but uh yeah actually yeah it's quite uh these aren't common okay um yeah i said what this i said this okay yeah so in summary we just make three assumptions we say ah some subsystem has lagrange dynamics it has a steady state distribution and it has a markov blanket and return we have identified an agent yeah nice sounds good okay so let me uh is there any question up to now there's a few more general ones but feel free to continue and we'll get them i think at the end because this next section will probably be very helpful okay so so now um let's talk about what the relation between this fep asian construction and the fep agent parts of the feb literature are according to the fvp literature uh so the these i kind of said this already but uh the thing that makes uh fep or active inference those are actually as far as i know agent special not two things or at least two things i mean there might be more but i think these are the main ones or the most important ones at least these fep agents that constructed the constructed fvp agents that i talked about before they can be seen as kind of a plausible or used they are used kind of as a plausible model in neuroscience and this is related to predictive coding and the patient brain but i actually i'm not qualified to talk about this because i don't know enough about neuroscience uh but the other thing that makes them special is the one of the main points i mean of this talk is that uh they are claimed to be based on first principles and when this statement that they are based on first principles actually means uh refers exactly to this this uh fep agent identification method so the first principles are basically what i just uh talked about that the larger equation and the steady state and the markov blanket gives you this agent so let's uh in order to make sure we're not putting words into the mouth of the literature let's read this quote once that's from also sophisticated inference paper that just came out the key distinction is that base adaptive

reinforcement learning considers arbitrary reward functions while sophisticated active inference optimizes an expected free energy that can be motivated from first principles okay actually here it doesn't say based but actually in another like you can look at this paper and um in another place it says based on first principles and also this is not only this is not only mentioned for the sophisticated active insurance agents you will see similar statements in the literature in other places so this is i mean this is kind of in some way it's super cool if it's true but i'm not so sure it's true um okay so some notes yeah okay so so this is about whether actually this how important is that is this how important is it for the constructed agents to be based on these first so in this also in the sophisticated inference paper the authors say that the agents expected free energy expected free energy which is the thing they minimize can be motivated from first place principles but they also seem to say that the expected free energy can also be i mean yeah actually i mean they say in in one place they say that it can also be motivated via information gain don't and they or i mean there's there's one part that this refers to so as far as i understand it they are there's a there are two ways to motivate this expected free energy minimization and one is via these first principles and one is via some other derivation that involves information now i wouldn't say i understand these derivations this direction completely but i do know that there's a paper by cohen and hotter and they show that if you that there's a that that basically some combination of information gain and reward maximization uh is you can prove that it's optimal in some sense so since this expected free energy is kind of a combination of those two things i kind of think let's put it this way i'm not surprised that it has it shows good performance or something like this because maybe it's actually so closely related to the method by cone that it just works so all i'm saying i guess what i want to say here is that in case the motivation via first fails uh the fep construction of agents may still be of interest and might still work and this also means that all these all publications i think there are some but yeah i think there may be quite a few all these publications that uh that are basically based on this expected free energy minimizing agents they they need not be without merit just because the first principle's motivation failed so um or in other words they can still they they can still make as much sense basically even if this uh free energy principle agent identification is wrong if i could ask a question here martin um yeah what what kind of first principles are you most interested in and then what kind of a theory of agents or agency would be satisfying given your interest in first principles yeah i mean i i i think so so if you see i'm interested in the first principles that work i don't know what the first principles are but like maybe one one first principle is that you can start from

physics like that's like you can you start with some well i don't know where five to go up yeah something like this you start with a dynamical system and you then derive using only notions that are well defined for dynamical systems or within such dynamical systems um to show that uh this thing this system contains agents or not or yeah so i guess the first principles for the first principles here are that okay all we have to believe is that there are such subsystems that obey the larger equation and i guess since the larger equation has been quite uh popular in physics um it has this has this this this has some strong support um and maybe that's sufficient to be your first principal uh also having steady state is fine it's well defined there's no i mean that's a mathematically well-defined thing we can in principle check this so yeah then the other thing that's used in the literature is the markov blanket and i think that's also yeah it's a it could be a well-defined thing um i think it's a bit of a like what it means in the fvp literature is kind of not it's not really clear and it changes and maybe has to change again but at least the original so if yeah it depends a bit um yeah i guess i have to i would have to show a bit more about the markov blanket but yeah i would say i yeah for first principles for me means probably just that you have to be able to define it for uh dynamic assist it has to be like all the notions you use you can only use notions that are defined for dynamical systems totally makes sense it's like if you're going to play the game of dynamical systems analysis you can only use those rules exactly you can only yeah exactly you have to you have to construct everything from there now you don't get anything else that's it you get the dynamical system uh and that's it yeah okay uh oh suspense boss yep that's where i was now yep i said this need not be there yep okay so now oh okay so that's uh a big part of the talk that's kind of that was kind of my personal my current personal view on what's yeah and how i yeah how i look at the fvp literature and what yeah how i split it up like i said and uh yeah so now i would go and talk about basically the issues i see with this uh fep agent identification method but if there's any questions now might be what's a good point that's it for now um definitely after this we'll see if anyone has any questions but feel free to type them in the live chat if anyone has it okay so the first issue i have is that this markov blanket assumption is known to be insufficient uh by now it's known to be insufficient maybe in the maybe at the for the very first paper it wasn't known to be insufficient or actually yeah for the first two papers maybe no yeah i don't know yeah yeah i can't really say because it wasn't me but yeah you know it's it's it's fair to say that uh the first two papers they might not have known this okay uh yeah so so so i'm gonna kind of go through the markov blanket evolution a little bit not not not in any detail but the mark of blanket evolution in the fep um maybe i should say like originally i did i think i don't

have this here uh i mentioned it down there okay so originally the the turmark of blanket comes from judea pearl from this 1988 book yeah so it's a it's a technical uh term that had a definition before it was used by the fap literature okay so and in this in this original paper the life as we know it there is actually no definition of like there's no no it's nowhere clear it's it's not clearly said what is meant what kind of markov blanket is meant i mean there's some kind of graph but markov blankets are come from directed acyclic grass and the graph is cyclic so it's actually not clear what is meant i mean there is but it's clearly stated that the markov blanket should imply this equation that the vector field when we split it up into the four coordinates also actually so it kind of implies also that there are these four we can split up the the coordinate into four sets of all the coordinates into four sets of corners and um the vector field then has can be uh has a certain shape form namely that the internal and sensory states do not um the the vector the vector field components of the internal and sensor sensory states do not directly depend on the internal states and the vector field components of the action and x action active and internal states do not depend directly on the external states maybe i said something wrong but yeah and we call this the so we we we call this the vector field condition or or yeah i mean i call this the vector field condition because it's a condition on the vector field and in our paper this is condition one okay so this is kind of i think in life as we know it this is kind of this is the only uh real play is where uh the markov blanket plays a role and by the way this is not a mark of blanket i mean not a technically this isn't a microphone i mean this isn't a pearlian markov blanket's condition um then in uh at least in in thomas past paper in 2020 the markov blanket is said to imply this condition the independence of the external states given the blanket states which are the sensory and the action states so we get this factorization of the stationary or steady state distribution and that's also considered a condition of the markov blanket this thing is also implicit in the this condition is also implicit in the free energy principle for particular physics but it's not explicitly referred to as the mark of naked so it's a bit weird but um this is actually a markov blanket in the sense of judy airport so this condition is a true mark of markov link and it has come to be called not only by me uh the time synchronous thinkerness markov blanket condition and in our paper this is condition two those two conditions this one and condition one they are independent of each other like you they are systems that obey this one and not this one and system the systems that obey this one and not this one and those systems are not complicated they are actually linear systems so they are these orange time own background processes where you can easily compute stationary distribution and it's a gaussian so you can look this up in the in the and i think i think one of

the i mean like this is wired speculation but um i think that uh the the authors or carl fristen because he's the only author uh kind of thought that this equate that this condition implies this condition i think it seems to me there are various reasons but it's kind of like i'm not going to go into it but yeah this seems to me the case and maybe you could think i mean i can like you could think this i mean it's not completely out of the question but yeah so then um we wrote we wrote this paper uh after we wrote this paper here in and complained about some some uh thing in the some things in the free energy principle for a particular physics namely this thing called the conditional independence corollary which is supposed to follow from the vector field condition i think um is the i mean i think that's the claim but we showed that it doesn't and so then the markov blanket or then there an an additional condition was uh presented in this paper called uh some interesting observations about the free energy principle or something this is the response to our paper and this condition is that the solenoidal component matrix r or q has this shape okay that's why i mean i'm going to call this the sony needle condition if i ever need it again in this talk but yeah since this came out after i mean this was in response to our paper so we don't really talk about this um okay so this is just some additional condition uh and then i think so yeah so you can kind of see all of these like these conditions are all used in to derive this statement here that that this statement here the fep agent identification method that you get to identify that we actually get we can identify an agent that actively acts 2 that that acts to uh persists basically with the goal of systems okay so we have this this condition as well and then in in par in the thomas past paper and also in the monograph if we look closely there is actually an additional assumption made um in the free energy principle for particular physics it's claimed to be a consequence of the other conditions but that's not true um uh we can show this it's kind of easy uh and in in the par paper it's it's uh actually said that it's an assumption so i've known that this is an additional assumption and this assumption is that there's some function σ and this function σ relates to variables and these two variables are the most likely external state both beta given the blanket states s_{na} and the most likely internal state given the blanket state s_{na} according to the stationary distribution the steady state distribution and the and so σ is uh the assumption is that that that there exists a function σ that maps one that maps the most likely internal state to the most likely external state um and that that this function doesn't need to exist you can see for example i mean it's it's not in our paper i think we should have probably put it in our paper but we didn't account example is uh you can use basically the system we pro we show in appendix d and then we show one of those linear linear systems again and for this the bold μ is just one half

a and the both eta is one half s and since s of t is not a function of a of t and basically uh so they are kind of so every combination of s and t can occur so there is no functional relation between the two and that means there also cannot be cannot be any function sigma um so so this means we kind of have to additionally assume sigma and also yeah yeah it and the thing is in my opinion this sigma might actually be a much stronger the strongest of all of these assumptions that are made like stronger than any of the other conditions here these other three conditions the vector field the tsnbc and the traditional and the solenoidal condition i think it does probably the most work but since i don't really understand the whole picture i can't really self-care for sure but so yeah we'll just start on that um sigma is sort of the proposed function that's literally linking internal and external states so when we're talking about agents and the way that their internal generative model is being linked to generative processes in the outside world that's like the key link that's what we're all about so i totally i totally see where you're coming from exactly that's a good yeah that's a good way to say it exactly i mean yeah i think i will i will say this later again but let me just say it here already these three conditions they are all trivially satisfied by four independent stochastic processes so like it seems uh kind of unrealistic let's put it this way that any of these conditions by themselves or even their combination would be sufficient to give us this kind of active active resisting to anything it's like or modeling of the environment or um asian inference about the environment four independent stochastic processes don't seem to i mean like it would be hard to argue that any of those is doing anything he's modeling any of the other ones so uh yeah exactly this is um but if you have such a then yeah then yeah there might be something going on okay so in summary we have these four conditions the vector field condition the t as mbc the sony recognition and the sigma function existence and i would say maybe call those the fep blanket together but okay okay here i say this now apart from the sigma function all the conditions are satisfied by four independent linear stochastic processes with gaussian white noise yeah so oh probably sigma is the most significant assumption and yeah i mean it's maybe a petty point i kind of think that calling the combination of these conditions markov is at least i mean by by now it is definitely misleading i mean i think the use of this word microfling it was misleading before but i think uh it should probably be dropped but anyway i'm gonna call it fep blanket oh sorry this is so and one more thing i am not i i cannot say that these conditions these four conditions together together are sufficient for uh the whole fep agent identification argument uh just because i don't really i haven't really gotten through the whole thing but we know um that they are necessary i mean they are used in uh arguments okay so that's the first that was

the first issue how am i doing in time actually it's like already super long you can go for as long as you want and we can go as long as you want already at one hour on twitch you can go as long as you want today martin and then we can always schedule another time to speak but this is really really helpful and i'm sure that a lot of people are gonna digest it and have a lot to talk to you about okay okay then i'm just gonna i'm just kind of taking okay all right next one uh this point is not in our paper um i guess it has been in the back of our heads but i mean yeah i think it's it's really also it's become much more i've become much more convinced about it and yeah so i'm gonna just mention it uh and i think the it would yeah i think it's a kind of challenge to the fep community maybe and i kind of there if i yeah i think they are kind of aware of this by now also so the thing is that there is no example for which all of the fep blanket conditions the four i just mentioned the four conditions i mentioned have been proven to hold so we don't know any longer an equation obeys all of these conditions oh actually i forgot the steady state maybe it should be but yeah there's no longer an equation with a steady state that obeys those so what this means is we don't even know if there's any system uh that the fep identification method applies to or that this statement that uh in very short that that the fvp blanket implies uh an agent that this is like we don't know that there's a system for which this is true so this also means we cannot we can never check like there's no system we could look at and check if there is actually something doing bayesian inference and in what way it's doing it and that it's actively resisting dispersion so this is kind of uh it's it's not there yeah i mean like it would be kind of cool to like so in some way this says the whole markov blanket story or yeah the whole as i call it fvp agent identification story might not might be vacuous like this is what this says and i i think it's kind of a serious point and it should be addressed i think okay so maybe some notes on this uh so there's this primary audio soup example in the original paper and it's used i mean the life as we know it and it's also used in the monograph but there's no proof that this thing has a steady state or that this steady state or base the gsmbc the time synchronous markov blanket um and also of course there's no proof that it has a sigma function because in life as we know it there wasn't even any sigma function also this uh like you could say maybe the constructed fep agents are systems that obey all these conditions but and then i mean maybe there are even some of them but this connection is never made explicitly like it's always just there's this claim that they are based on this uh on these first principles and this all of this but um it's never shown for example that the agent environment combination which forms this markov process that it has a steady state distribution and it's actually very unlikely that in all of them it has a steady and actually you can kind of like i will

argue that it's actually not true definitely not for all of them and also even if uh this agent environment pair would have a steady state distribution we would still want to see that it obeys the tsabc and also that is kind of question so i should mention that this this is like the second point that checking that the tsabc holds uh is an idea that's by nathaniel virgo and uh yeah there's some arguments about it but i won't go further into it the so furthermore it is never checked like we never look at the sigma function which appears to place a significant role but we never look at this signal function in these constructed uh fvp agents um what what is kind of what i'm pretty sure that it says is satisfied in a reasonable way is the vector field condition yeah and maybe this i don't really know why i say this here yeah like the ah okay so you could you could yeah okay so so you could maybe so since me and felix we use the launch onscreen openback processes to show various uh things about um the free energy principle and we we know that it satisfies quite a few of like it satisfies uh three of these assumptions so it satisfies the vector field condition we can make it so that it satisfies the tsabc and this solenoidal condition but there is no such system like so basically i'm like if you want to there's no such system that has a sigma function so if you wanted to search for an einstein openback process that obeys that is basically an existence proof then i can tell you it's impossible so this proof is kind of unpublished but uh and it's uh mostly due to fedex but uh yeah maybe maybe i'm gonna maybe you're gonna put it on archive at some point it's really not complicated um okay so that's this point now i want to make also another point that is not in our paper and then we're going to slowly get to the end because the last point isn't very substantial okay so yeah what i want to say is that there are agents in the fep literature the system that they are contained in so if we form the environment if we look at the environment and the agent in combination as a markov process what i talked about in the very beginning then and and the the agent is this uh fep then for some of them it's straightforward to show that they that combination this combine this resulting markov process does not have a steady state and the first thing to note is that if you have some kind of some monotonously increasing counter it can never so if your state contains a mono monotonously increasing counter it means you can never return to a previous state and this means you cannot have a state because the states are not recurrent and that means there is no steady state so as i mentioned before the internal state the internal agent state of uh fep agents are hyper parameters and are updated uh using bayesian inference variation of bayesian inference but that includes exact patient inference and so these hyper parameters of x exact they accumulate evidence which means they count occurrences of events and they only count up and they count every and for every event they

count something so they are counters that always monotonously increase and for example in the sophisticated inference paper days if you look at 4.5 there is one there is an explicit equation for the updating of one of the parameters which counts the occurrences of transitions from expected environment states to observations and no such counter can be contained within any Markov chain that has a steady state so this means that this agent this particular agent in the sophisticated inference paper is not within such a system that has a steady state so it's not within a system that a subsystem it cannot exist within a subsystem that obeys the conditions of the FAP agent identification okay so uh this so be aware that this statement also is not published uh but i kind of want to put it out there because it would be boring if i just talk about that's already published and um so but they i mean like there's some caution advice what i think but if you want to look at this at a very simple example of this uh i published a paper last year on this um and so i'm pretty sure and this it seems pretty straightforward this argument and then one more thing there so in this paper sophisticated inference also there is some argument about steady state in appendix A2 but that steady state just kind of just it ignores the internal state of the agent it only looks at the other variables so i think and i think that's actually necessary like there's just no steady state if you include the internal state of the agent so maybe be aware of this and uh yeah the two consequences of this are um FEP agents are not identified by the FEP agent method and the and this also means that the FEP agents cannot be motivated by this uh FEP agent identification or or by first principles because they actually violate these first principles or in this case at least this particular agent okay uh i think that was yeah that was this point the last point is kind of uh a bit nitpicky but i think it's also kind of important for some people and that is that this uh identification is changing in the literature and you shouldn't think for example that the original life as we know it a paper contains a method that's very similar to the neurons i mean the these constructions of the most likely internal and external states they don't appear there is no sigma but in the later versions these things these constructions play essential roles so these two they are two completely different methods in okay and and yeah one of these things we did in our paper is we showed that this original method in life as we know it is kind of wrong but oh wow where did i go this was end so okay don't worry we're not gonna look at all these slides uh okay here we are okay so we we kind of looked at this and we think it's it's wrong uh at least the way it stated um and yeah okay so so be aware that if you look at this method like don't maybe maybe just ignore the life as we know in paper that's maybe what i want to say and also like you can't tell from the literature that this method is uh very different i mean it's

kind of they never say why they changed it and it's kind of still repre like still mentioned as the basis of these things so be aware it's kind of maybe it's a waste of time reading this okay that's it uh in summary what i said is that we can separate the literature into agent identification fep agent identification and afep agent and that the fep agent identification requires significantly more assumptions than just whatever the mark of blanket was in the beginning uh that this uh agent identification method misses an example that would prove that it that such systems actually exist such systems with agents that systems with such agents actually exist and it requires a steady state that at least some of the fvp agents definitely don't satisfy the method has changed from its original but also note it could still be correct i don't know like maybe there are enough assumptions you can make and maybe they are still systems that contain such asians uh it could all still work out but i haven't made it through all of these derivations so they may be like i said there may be even more necessary okay and the last point that fep agent construction the fvp agent construction is probably unaffected by any of these problems of the fep agent identification method or the mark of blanket stuff if you want okay and it can be yeah okay and this can be probably independently motivated by some other arguments that are not first principles if you like okay that's it i thank a lot of people for talking to me about this stuff uh yeah and that's it thank you very much for your attention if you made it all the way especially if you made it all the way thank you martin maybe you can unshare and i'll just ask a few of the questions from the chat in closing okay it's is this the where is the unshare is it the same as um maybe go back to the jitsi and just unshare it yeah cool it's great great nice cool okay perfect so great um thanks again that was really awesome and i hope people paid attention so the first question is peter and peter asked the distinction between agent identification and agent construction is very nice if i wanted to use that distinction in my own work should i cite your paper or is this distinction made elsewhere too i'm wondering if i if i mention this anywhere i don't know actually uh i don't know i mean like it's in some way it's just yeah it's kind of like a yeah it's kind of like an observation that makes a lot of sense and it's been yeah i don't know no i like if i if i think of something maybe i don't know can i see you are you on twitter i don't know uh maybe i'm gonna post it on twitter follow me on twitter i think it's it's kind of an obvious thing uh and i'm not the only one who has uh i'm definitely not the only one who has uh observed this i yeah probably there's somebody in the did it in 1950 but yeah i don't know like i'm not going to be mad at it's you if you don't cite me let's put it this way thanks for the response another question from peter is how much should the agent and agent construction depend on each other in other words how big is the problem if they

don't and how important is it that it comes from first anyways can i say the first sentence again i'm sorry uh how much should the agent identification and agent construction depend on each other okay so are those like two separate lines or one integrated hole yeah good question i mean i guess i mean in i guess ideally there would be some kind of relation but uh on the other hand so we want to construct agents like so so i in my opinion the bacteria are agents but when we do agent construction we don't really care like we don't have to follow the principles of bacteria necessarily if we don't like either we have i mean we might be we might have to because that's just how physics works or due to the constraints of physics but if we don't have to then nothing should stop us just constructing any kind of thing that we want i mean like like if we uh yeah so i guess they are not necessarily dependent i mean yeah at least i can't i can see this so on the other hand i guess it would be it would also be nice if you have some yeah if you still have some agent identification method which comprises all possible agents like all agents that you construct should be uh examples of your like should be detected or identifiable by your agent identification method so okay now i'm gonna take back what i said before they should be independent i hope that i hope this helped a little bit yep and and i'll just give one i think there's probably more to say but that i can't at the moment i can't really think of it so here's one final question and is a question i had but i'll use the phrasing peter here and so this will be our closing question hopefully the beginning of something new in the beginning of moving forward peter wrote i suppose the mathematics in free energy principle and active inference could be taken as a work in progress are there suggestions for how to solve or work with the issues that you're raising just any thoughts on this could be nice so where do we go as learners and doers practitioners and communicators in active inference so so yeah like i said i mean on the one hand you can if you if you construct agents you can just ignore what all of this stuff i said because in my opinion you're not relying on this agent identification method anyway so so there's no problem like of course you could maybe wonder what this base optimal that is sometimes mentioned in the literature of the fep what this really means uh and where it comes from like i think that's i find it i always think it's a little bit vague what this really means and it's actually not such a trivial question what base optimize but or what optimize um but yeah mostly you can ignore this fep identification but if you are interested in the fep identification then yeah i mean that i guess the the the thing is look at uh either i mean try to understand where it's actually going i mean like i still haven't really understood these later parts of the the monograph where everything is supposed to come together because i always get stuck um before in some equations that i don't understand and uh then i think

they're wrong but i'm not sure they're wrong yet as i would have mentioned that but there's some way i don't i'm i suspect they're wrong but yeah i'm still not sure and but yeah i guess that's still like i don't really see like if you want to save something of this method then look at another another other way is maybe to think about uh think about this for yourself think about what it what are the conditions you think under which something is doing asian inference or something like this and ignore all the math by then that's already in the literature and then try like once you figured it out for yourself maybe uh try to come back to the literature and compare it or something i sometimes think people rely too much on what this what the literature says and i'm not really sure that actually what's in the literature is like it doesn't strike me as a very straightforward argument um and i like straightforward arguments i think straightforward arguments are usually better um like maybe there isn't a straightforward argument but i kind of think i always think there's a straightforward argument um but that's just my opinion yeah it's off well it's a nice point that the literature is a scaffold for our own thinking and development the literature is not uh a road that's been paved with gold and it's just perfect in the past it's actually always a work in progress and always a scaffold so thanks so much for sharing this really work in progress not that it isn't uh with landmarks that you've published but it is a work in progress in the bigger sense so also thanks for staying up late to bring us this information and we hope that in future live streams unpack this more and hear updates as you continue to progress on this line of research okay yeah thank you very much for having me thank you martin peace all right bye peace